

## Quiz (Carolina Reaper)

**Q1.** Solve the inequality  $|3x - 2| > 5$ .

- (A)  $x < -1$  or  $x > 3$   
 (A)  $x > -1$  or  $x < 3$   
 (A)  $x > 1$  or  $x < -1$   
 (A)  $x < 1$  or  $x > -1$   
 (A)  $x > 2$  or  $x < -2$

**Q2.** A delivery truck travels  $|x - 100|$  miles in  $x$  hours. If the truck's speed is between 40 and 60 mph, which inequality represents this situation?

- (A)  $|x - 100| \geq 40x$   
 (B)  $|x - 100| \leq 60x$   
 (C)  $40 \leq |x - 100| \leq 60$   
 (D)  $40x \leq |x - 100| \leq 60x$   
 (E)  $|x - 100| = 50x$

**Q3.** The absolute value inequality  $|2x + 5| \leq 3$  is equivalent to

- (A)  $-4 \leq x \leq 1$   
 (B)  $-1 \leq x \leq 4$   
 (C)  $-2 \leq x \leq 3$   
 (D)  $-3 \leq x \leq 2$   
 (E)  $-1 \leq x \leq -1$

**Q4.** Graph the solution set of  $|x + 2| > 4$  on a number line.

- (A)  $(-\infty, -6) \cup (2, \infty)$   
 (B)  $(-\infty, -2) \cup (6, \infty)$   
 (C)  $(-6, 2)$   
 (D)  $(-2, 6)$   
 (E)  $(-\infty, -2] \cup [6, \infty)$

**Q5.** Express the solution set of  $|3x - 1| \geq 2$  in interval notation.

- (A)  $(-\infty, -1/3] \cup [1, \infty)$   
 (B)  $(-\infty, 1] \cup [1/3, \infty)$   
 (C)  $(-\infty, 1/3] \cup [1, \infty)$   
 (D)  $(-\infty, -1/3] \cup [1/3, \infty)$   
 (E)  $[1/3, 1]$

**Q6.** A shipping company charges  $|x - 50|$  dollars for packages weighing  $x$  pounds. If the charge is between 10 and 20, which inequality represents this situation?

- (A)  $40 \leq x \leq 60$   
 (B)  $30 \leq x \leq 70$   
 (C)  $10 \leq |x - 50| \leq 20$   
 (D)  $40 < x < 60$   
 (E)  $|x - 50| > 10$

**Q7.** Solve the inequality  $|2x + 1| < 7$ .

- (A)  $-2 < x < 3$   
 (B)  $-4 < x < 2$   
 (C)  $-3 < x < 1$   
 (D)  $-1 < x < 3$   
 (E)  $-2 < x < 2$

**Q8.** The graph of  $|x - 3| = 2$  is

- (A) two points:  $x = 1$  and  $x = 5$   
 (B) one point:  $x = 3$   
 (C) a line segment:  $[1, 5]$   
 (D) a vertical line:  $x = 3$   
 (E) the  $x$ -axis

## Open Ended Questions (FRQ)

**Q9.** A traffic flow model is given by  $|x - 20| \leq 5t$ , where  $x$  is the distance traveled in miles and  $t$  is the time in hours. Find the distance traveled after 2 hours and express your answer in interval notation.**Q10.** The delivery time  $t$  in hours for a package is given by  $|t - 3| \geq 1$ . Find the values of  $t$  for which the delivery time is between 1 and 4 hours and express your answer in interval notation.

## Chef's Tips

- Read the questions carefully
- Use a ruler for graphs
- Check your units